Lecture Notes
EC226 Term 2 2024/25

Cointegration

Reading:

- Stock and Watson (2003): Chapter 16
- Dougherty (2016): Chapter 13
- Wooldridge (2013): Chapter 18

Contents

1 Definition

The variables y_t and x_t are said to be cointegrated of order d, b, if

1. the variables y_t and x_t are I(d) (that is, they need d differences to induce stationarity)
2. there exists a vector β ($\neq 0$) so that $\varepsilon_t = y_t - \beta x_t \sim I(d-b)$

The coefficient β is called the cointegrating vector (the coefficient parameter in the case of there just be two variables). In economics the case generally considered is when $d = b = 1$. This is an important result as any arbitrary linear combination of I(1) series will be I(1) (unless the series are cointegrated).

NOTE:

You cannot find cointegration between

1. an I(0) (stationary) dependent variable and I(1) (non-stationary) explanatory variables
2. an I(1) dependent variable and I(0) explanatory variables - the regression is unbalanced.
3. an I(0) dependent variable and I(0) explanatory variables - the regression is unbalanced.

. For the series to be cointegrated they must have a **common I(1) (stochastic trend) component**, such that a linear combination of the series eliminates this common stochastic trend.

Consider, for example,

$$x_t = \sum_{j=1}^{t} \xi_t + \varepsilon_t = z_t + \varepsilon_t$$

and x_t is made up of two elements:

1. the I(1) component, (which is the stochastic trend) $z_t = \sum_j \xi_j$
2. a stationary I(0) component, ε_t

If we also have:

$$y_t = \beta z_t + \varsigma_t$$

then y_t has the same I(1) stochastic trend component, z_t, as well as a distinct stationary term, ς_t.

There is a linear combination of y_t and x_t will eliminate the common stochastic trend, z_t, and this is written as:

$$y_t - \beta x_t = (\beta z_t + \varsigma) - \beta(z_t + \varepsilon_t) = \underbrace{\varsigma_t - \beta\varepsilon_t}_{\epsilon_t}$$

the term $\varsigma_t - \beta\varepsilon_t$ is a linear combination of two I(0) series and is therefore I(0) [1]. This means that while y_t and x_t are both I(1), there is a linear combination $y_t - \beta x_t$ which is I(0) and this is an example of cointegration.

One can think of cointegrating equations as long-run "equilibrium" equations between economic variables. An alternative approach to discovering the relationship between I(1) series is to make all of the series I(0) series (by taking a first difference of the series) and then to look for linear relationships between these I(0) series. However, this approach throws useful long-run information away as it is solely concerned with short-run movements, ie. the value of a series today, y_t, compared to its previous period y_{t-1}.

[1] Any linear combination of stationary (I(0)) series must itself be stationary (I(0))

2 Spurious vs cointegrating relationships

In trying to find equilibrium (long-run) relationships between I(1) series, one needs to be careful to distinguish cointegrating relationships from non-cointegrating relationships (known as spurious relations).

2.1 Spurious regressions

Consider the case, where you have a variable, y_t

$$y_t = y_{t-1} + \varepsilon_t \text{ where } \varepsilon_t \sim (0, \sigma_1^2) \Rightarrow \quad y_t = y_0 + \sum_{j=1}^{t} \varepsilon_j$$

and a variable, x_t

$$x_t = x_{t-1} + \varsigma_t \text{ where } \varsigma_{2t} \sim (0, \sigma_2^2) \Rightarrow x_t = x_0 + \sum_{j=1}^{t} \varsigma_j$$

In this case both y_t and x_t are random walk processes without drift (and so are I(1) series). If we also assume $E(\varepsilon_t \varsigma_t) = 0$, then there is nothing linking the two series together. Despite this as y_t and x_t are trending over time (this is because of the stochastic trend $\sum_{j=1}^{t} \varepsilon_j$ in y_t and $\sum_{j=1}^{t} \varsigma_j$ in x_t), a regression between the two variables:

$$y_t = \alpha + \beta x_t + \varepsilon_t \tag{1}$$

we reject $H_0 : \beta = 0$ (which is true) around 77% of the time (for a 5% significance level). If y_t and x_t were nonstationary processes with a non-zero drift parameter these rejection rates would be even greater.
Consequently, we need to distinguish between

1. Spurious (nonsense) regressions, in which we observe an apparently significant relationship between unrelated series.

2. Genuine cointegrating relationships between series.

2.2 Cointegrating regressions

For ease of notation, we will consider testing for cointegration when we have two ($n = 2$) I(1) variables, in which case there is at most $r = 1$ cointegrating relationship. The analysis can be extended to looking for cointegration between $n \geq 2$ variables, in which case there could in principle be up to $r = n - 1$ cointegrating relationships - see Johansen (1988) for looking for multiple cointegration relationships. In this module we will limit our analysis to identifying just one cointegrating relationship using the approach of Engle and Granger

(1987), who recommend a two-step procedure for cointegration analysis (between up to k variables):

STEP ONE

Estimate a long-run (equilibrium) equation either by:

1. Estimating a static (long-run) equation such as:

$$y_t = \delta_0 + \delta_1 x_t + \varepsilon_t \tag{2}$$

   by OLS, or

2. Estimating a dynamic equation, such as:

$$y_t = \gamma_0 + \gamma_1 x_t + \gamma_2 x_{t-1} + \gamma_3 x_{t-2} + \gamma_4 y_{t-1} + \gamma_5 y_{t-2} + u_t$$

   by OLS, and solving for the long-run equation, that is:

$$y_t = \frac{\gamma_0}{(1-\gamma_4-\gamma_5)} + \frac{(\gamma_1+\gamma_2+\gamma_3)}{(1-\gamma_4-\gamma_5)} x_t + \underbrace{\frac{u_t}{(1-\gamma_4-\gamma_5)}}_{e_t} \tag{3}$$

Either equation (2) (or equation (3)) is the long-run equation and the residuals from equation (2), denoted as:

$$e_t = y_t - \hat{\delta}_0 - \hat{\delta}_1 x_t \tag{4}$$

Where $\hat{\delta}_j$ $j = 0, 1$ is the OLS estimator of δ_j $j = 0, 1$, or the residuals from equation (3), denoted as:

$$e_t = y_t - \left(\frac{\hat{\gamma}_0}{(1-\hat{\gamma}_4-\hat{\gamma}_5)} + \frac{(\hat{\gamma}_1+\hat{\gamma}_2+\hat{\gamma}_3)}{(1-\hat{\gamma}_4-\hat{\gamma}_5)} x_t \right) \tag{5}$$

are measures of "disequilibrium" between y_t and x_t and cointegration requires that these residuals are stationary.

The stationarity of the residuals e_t, is determined using an ADF test as

$$\Delta e_t = \mu + \gamma e_{t-1} + \sum_{j=1}^{p} \delta_j \Delta e_{t-j} + \eta_t$$

and we are testing $H_0 : \gamma = 0$ (the residuals are non-stationary and therefore the regression is spurious) versus $H_1 : \gamma < 0$ the residuals are stationary and therefore the regression is a cointegrating regression.

The critical values are taken from MacKinnon (1991) have to be corrected for the number of non-stationary variable you estimated in the long-run equation, n. The reason we need to adjust the critical value is because OLS finds the parameters which minimise the RSS (and hence the finds the residuals which are most stationary), which therefore biases the ADF test to finding a cointegrating equation, consequently we adjust the critical values for n the

number of non-stationary variables in the long-run equation.
STEP TWO:
If there is a cointegrating equation then you should estimate an Error Correction Model (ECM) of the form:

$$\Delta y_t = \phi_0 + \sum_{j=1} \phi_j \Delta y_{t-j} + \sum_{h=1} \theta_h \Delta x_{t-h} + \alpha e_{t-1} + \eta_t \tag{6}$$

by OLS, where we assume η_t is a well-behaved error term, as this equation has only I(0) variables (remember the long-run residuals e_{t-1} are stationary and all other variables are in first differences) standard hypothesis testing using t-ratios and diagnostic testing of the error term is appropriate, note $\alpha < 0$ as this is an error correcting model, such that if:
$e_{t-1} > 0 \Rightarrow y_{t-1} > \hat{\delta}_0 + \hat{\delta}_1 x_{t-1}$
and therefore y_{t-1} is above its equilibrium value and $\alpha e_{t-1} < 0 \Rightarrow \Delta y_t < 0$, meaning that the dependent variable falls.
Similarly, if
$e_{t-1} < 0 \Rightarrow y_{t-1} < \hat{\delta}_0 + \hat{\delta}_1 x_{t-1}$
and therefore y_{t-1} is below its equilibrium value and $\alpha e_{t-1} > 0 \Rightarrow \Delta y_t > 0$, meaning that the dependent variable rises.
In general the dependent variable is adjusting (correcting) for disequilibrium (errors) in the previous period.

2.3 Non-cointegrating regressions

If you do not have cointegration in **STEP 1** then you do not have an equilibrium equation and therefore there you cannot build an error correction model as there is no long-run (equilibrium) errors for which the model needs to correct. The short-run equation should still be balanced and so we still need to difference all of the non-stationary variables to make them stationary:

$$\Delta y_t = \phi_0 + \sum_{j=1} \phi_j \Delta y_{t-j} + \sum_{h=1} \theta_h \Delta x_{t-h} + \eta_t$$

This equation looks a bit like the ECM in equation (**??**), apart from (and very importantly) we do not have the error correction term, e_{t-1}, but we have still transformed all of the variables to be I(0).
If you have a combination of I(1) and I(0) variables in your regression and you do not have cointegration between the I(1) variables, then any short-run regression you estimate between the variables must involve only the I(0) variables (thereby ensuring the equation is balanced), and so we may have:

$$\Delta y_t = \phi_0 + \sum_{j=1} \phi_j \Delta y_{t-j} + \sum_{h=1} \theta_h \Delta x_{1t-h} + \sum_{h=1} \kappa_h x_{2t-h} + \eta_t$$

if

$$y_t \sim I(1),\ x_{1t} \sim I(1),\ x_{2t} \sim I(0)$$

OR:

$$y_t = \phi_0 + \sum_{j=1} \phi_j y_{t-j} + \sum_{h=1} \theta_h \Delta x_{1t-h} + \sum_{h=1} \kappa_h \Delta x_{2t-h} + \eta_t$$

$$y_t \sim I(0),\ x_{1t} \sim I(1),\ x_{2t} \sim I(1)$$

2.4 Super Consistency

In Term 1 we spent time worrying about whether biases in our OLS coefficients and found that in a number of instances the OLS coefficient are biased (and even inconsistent) due to

1. Omitted relevant variables
2. incorrect function form
3. measurement errors in the explanatory variables
4. endogeneity

However, we do not have to worry about these issues as much with cointegration analysis and estimating our long-run equations by OLS. The reason for this is down to the fact that the OLS estimators of the long-run coefficients are said to be "super-consistent".
The idea of super-consistent is that the OLS estimator approaches the true coefficient very quickly as the number of observations T increases. Consider the OLS estimator for equation (2) is:

$$\hat{\delta}_1 = \frac{\sum_t (x_t - \bar{x})(y_t - \bar{y})}{\sum_t (x_t - \bar{x})^2} = \delta_1 + \frac{\sum_t (x_t - \bar{x})\varepsilon_t}{\sum_t (x_t - \bar{x})^2} \Rightarrow \hat{\delta}_1 - \delta_1 = \frac{\sum_t (x_t - \bar{x})\varepsilon_t / T}{\sum_t (x_t - \bar{x})^2 / T} \tag{7}$$

now $\sum_t (x_t - \bar{x})\varepsilon_t / T$, is the covariance between a stationary I(0) series ε_t and the nonstationary I(1) series, x_t, and this term will not increase over time. In contrast $\sum_t (x_t - \bar{x})^2 / T$ is the variance of the I(1) series, x_t, and we know for I(1) series $V(x_t)$ increases with T. Consequently, as $T \to \infty$ the denominator becomes very large relative to the numerator, therefore the term

$$\frac{\sum_t (x_t - \bar{x})\varepsilon_t / T}{\sum_t (x_t - \bar{x})^2 / T} \to 0$$

meaning the bias disappears very quickly and this will always be the case providing the terms entering into ε are I(0) terms.
For example, consider the case where there is an omitted relevant variables (which is expressed as an incorrectly specified dynamic equation) and there is a true (T) equation and

an estimated (F) equation:

$$y_t = \beta_0 + \beta_1 x_{t-1} + \varepsilon_t \tag{T}$$
$$y_t = \delta_0 + \delta_1 x_t - \delta_1(x_t - x_{t-1}) + \varepsilon_t \tag{F}$$

Here the omitted variable is $z_t = -\delta(x_t - x_{t-1})$, which is an I(0) process. Therefore,

$$\begin{aligned} E(\hat{\delta}_1) =& \beta_1 - \delta_1 \frac{cov(x_t, z_t)}{var(x_t)} \\ &= \beta_1 - \delta_1 \frac{cov(x_t, \Delta x_t)}{var(x_t)} \\ &= \beta_1 - \delta_1 \frac{cov(x_t, \Delta x_t)/T}{var(x_t)/T} \end{aligned} \tag{8}$$

The numerator of the bias term in equation (8) will not increase with T as it is the covariance between an I(1) and an I(0) series, while the denominator will increase with T, therefore asymptotically the bias term goes to zero. The same proof works for measurement error, incorrect function form, and endogeneity bias, which are all likely to be I(0) biases.

NOTE:

The t-ratios from estimating equation (2) cannot be interpreted, as this long-run equation will have serial correlation (due to incorrectly specified dynamics) as well as omitted variable problems. The traditional diagnostic tests from (2) are largely unimportant as the only important question is the stationarity or otherwise of the residuals.

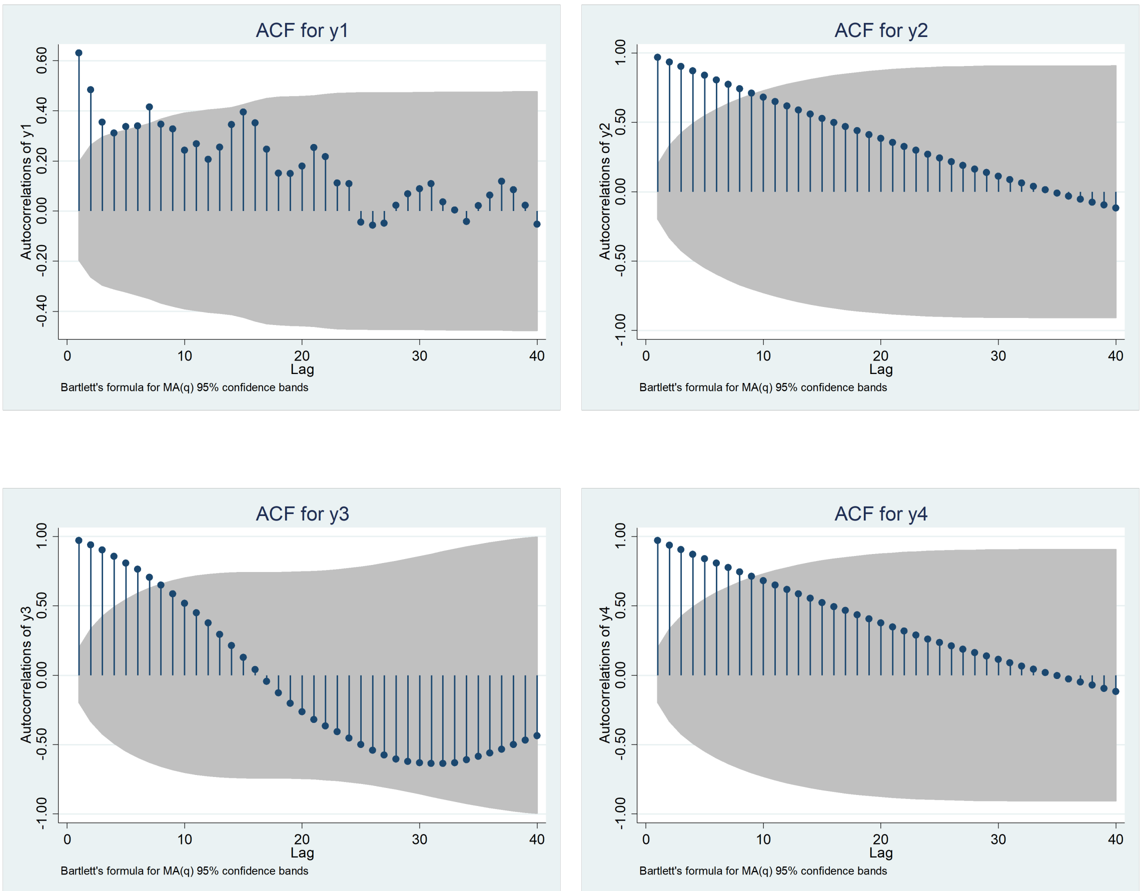
Appendices

A Example of Cointegration Analysis

Consider the 4 series y_{1t}, y_{2t}, y_{3t}, y_{4t} for which we plot the series and the ACF

Figure A1.1 Plot of the 4 series

Figure A1.2: ACF of the 4 series



For all series these is some evidence of potential non-stationarity and as a result we undertake unit root tests on each of the series using model C and 2 lags as the model, i.e. $\Delta y_t = \alpha + \mu t + \gamma y_{t-1} + \sum_{j=1}^{2} \delta_j \Delta y_{t-j} + \varepsilon_t$ With $H_0 : \gamma = 0;\ H_1 : \gamma < 0$

```
. dfuller y1 if index>10 & index<=110, trend lags(2)

Augmented Dickey-Fuller test for unit root          Number of obs   =       100

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -5.306            -4.040            -3.450            -3.150
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0001
```

Which suggest we are able to reject H_0 and $y_{1t} \sim I(0)$

```
. dfuller y2 if index>10 & index<=110, trend lags(2)

Augmented Dickey-Fuller test for unit root          Number of obs   =       100

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -0.878            -4.040            -3.450            -3.150
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.9584
```

Which suggest we are unable to reject H_0 and $y_{2t} \sim I(1)$

```
. dfuller y3 if index>10 & index<=110, trend lags(2)

Augmented Dickey-Fuller test for unit root          Number of obs   =       100

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -1.448            -4.040            -3.450            -3.150
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.8462
```

Which suggest we are unable to reject H_0 and $y_{3t} \sim I(1)$

```
. dfuller y4 if index>10 & index<=110, trend lags(2)

Augmented Dickey-Fuller test for unit root         Number of obs   =       100

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -0.734            -4.040            -3.450            -3.150
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.9708
```

Which suggest we are unable to reject H_0 and $y_{4t} \sim I(1)$.
Now we look for cointegration between pairs of variable to see whether they are cointegrated. Here we look at $y_{2t} \sim I(1)$ and $y_{1t} \sim I(0)$, so they are integrated of different orders and therefore cointegration make no sense.

```
. reg y2 y1 if index>10 & index<=110

      Source |       SS       df       MS              Number of obs =     100
-------------+------------------------------           F(  1,    98) =   55.40
       Model |  95756426.9     1  95756426.9           Prob > F      =  0.0000
    Residual |   169403142    98  1728603.49           R-squared     =  0.3611
-------------+------------------------------           Adj R-squared =  0.3546
       Total |   265159569    99  2678379.49           Root MSE      =  1314.8

------------------------------------------------------------------------------
          y2 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
          y1 |   81.11981    10.8991     7.44   0.000      59.4909    102.7487
       _cons |   94.46135   272.0408     0.35   0.729    -445.3948    634.3175
------------------------------------------------------------------------------
```

There is a very significant and positive relationship between the two, but if we undertake a unit root test on the residuals (denoted, res_y2):

```
. predict res_y2, resid

. dfuller res_y2 if index>10 & index<=110, lags(2)

Augmented Dickey-Fuller test for unit root         Number of obs   =       100

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -1.664            -3.510            -2.890            -2.580
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.4500
```

The critical value comes from MacKinnon's tables with $n = 2$ (see Appendix 2) and at the 5% significance level this is approximately -3.34 and we are unable to reject H_0. This is not surprising as the stochastic trend in y_{2t} cannot be cancelled by anything in the stationary series, y_{1t}, which does not have a stochastic trend – by definition.
Now we look for cointegration between y_{1t} and y_{2t}.

```
. reg y1 y2 if index>10 & index<=110

      Source |       SS       df       MS              Number of obs =     100
-------------+------------------------------           F(  1,    98) =   55.40
       Model |  5255.02379     1  5255.02379           Prob > F      =  0.0000
    Residual |  9296.68713    98  94.8641544           R-squared     =  0.3611
-------------+------------------------------           Adj R-squared =  0.3546
       Total |  14551.7109    99  146.986979           Root MSE      =  9.7398

------------------------------------------------------------------------------
          y1 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
          y2 |   .0044518   .0005981     7.44   0.000     .0032648    .0056388
       _cons |    13.5397   1.481803     9.14   0.000     10.59911    16.48029
------------------------------------------------------------------------------
```

There is a very significant and positive relationship between the two, but if we undertake a unit root test on the residuals (denoted, res_y1):

```
. predict res_y1, resid

. dfuller res_y1 if index>10 & index<=110, lags(2)

Augmented Dickey-Fuller test for unit root         Number of obs   =       100

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -4.980            -3.510            -2.890            -2.580
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000
```

The critical value comes from MacKinnon's tables with $n = 2$ and at the 5% significance level this is approximately -3.34. Therefore we are able to reject H_0. However, this test makes no sense as the dependent variable y_{1t} is stationary and therefore any residuals from this equation MUST be stationary – but this has nothing to do with cointegration. Now we look for cointegration between y_{3t} and y_{2t}.

```
. reg y3 y2 if index>10 & index<=110

      Source |       SS       df       MS              Number of obs =     100
-------------+------------------------------           F(  1,    98) =    2.02
       Model |  894.168856     1  894.168856           Prob > F      =  0.1586
    Residual |  43409.8336    98  442.957485           R-squared     =  0.0202
-------------+------------------------------           Adj R-squared =  0.0102
       Total |  44304.0024    99  447.515176           Root MSE      =  21.047

------------------------------------------------------------------------------
          y3 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
          y2 |   .0018364   .0012925     1.42   0.159    -.0007286    .0044013
       _cons |  -16.06086   3.201994    -5.02   0.000    -22.41511   -9.706602
------------------------------------------------------------------------------
```

There is no real relationship between the two series and if we undertake a unit root test on the residuals (denoted, res_y3):

```
. predict res_y3 if index>10 & index<=110, resid

. dfuller res_y3 if index>10 & index<=110, lags(2)

Augmented Dickey-Fuller test for unit root         Number of obs   =        97

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -1.456            -3.514            -2.892            -2.581
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.5552
```

The critical value comes from MacKinnon's tables with $n = 2$ and at the 5% significance level this is approximately -3.34 we are unable to reject H_0 there is no cointegrating equation and so there is no cointegration, but there is no evidence of a spurious regression either as the coefficient in the long-run equation is insignificant.

Now we look for cointegration between y_{4t} and y_{2t}.

```
. reg y4 y2 if index>10 & index<=110

      Source |       SS       df       MS              Number of obs =     100
-------------+------------------------------           F(  1,    98) =25991.32
       Model |  28028079.6     1  28028079.6           Prob > F      =  0.0000
    Residual |  105679.563    98  1078.36288           R-squared     =  0.9962
-------------+------------------------------           Adj R-squared =  0.9962
       Total |  28133759.2    99  284179.386           Root MSE      =  32.838

------------------------------------------------------------------------------
          y4 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
          y2 |   .3251195   .0020166   161.22   0.000     .3211175    .3291215
       _cons |   8.211186   4.995995     1.64   0.103    -1.703203    18.12557
------------------------------------------------------------------------------
```

There is a very significant and positive relationship between the two, but if we undertake a unit root test on the residuals (denoted, res_y4)

```
. predict res_y4 if index>10 & index<=110, resid

. dfuller res_y4 if index>10 & index<=110, lags(2)

Augmented Dickey-Fuller test for unit root         Number of obs   =        97

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -4.028            -3.514            -2.892            -2.581
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0013
```

The critical value comes from MacKinnon's tables with $n = 2$ and at the 5% significance level this is approximately -3.34. Therefore we are able to reject H_0 and this implies we have a cointegrating equation between the two series.

Given a cointegrating equation, we now look at building an ECM and the form of the ECM should be:

$\Delta y_{4t} = \beta_0 + \beta_1 \Delta y_{4t-1} + \beta_2 \Delta y_{4t-2} + \beta_3 \Delta y_{2t-1} + \beta_4 \Delta y_{2t-2} + \beta_5 \hat{u}_{t-1} + \varepsilon_t$

Where $\hat{u}_t = res\_y4_t = y_{4t} - 8.211 - 0.3251 y_{2t}$

```
. reg d.y4 l(1/2).d.y4 l(1/2).d.y2 l.res_y4 if index>10 & index<=110

      Source |       SS       df       MS              Number of obs =      97
-------------+------------------------------           F(  5,    93) =   16.82
       Model |  31173.9543     5  6234.79087           Prob > F      =  0.0000
    Residual |  34468.5311    91  370.629366           R-squared     =  0.4749
-------------+------------------------------           Adj R-squared =  0.4467
       Total |  65642.4854    96  669.82128            Root MSE      =  19.252

------------------------------------------------------------------------------
        D.y4 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
          y4 |
         LD. |    .481929   .0927188     5.20   0.000     .2978078    .6660501
        L2D. |  -.1372576   .1031376    -1.33   0.187    -.3420684    .0675531
             |
          y2 |
         LD. |   .5789762   .3857555     1.50   0.137    -.1870577     1.34501
        L2D. |  -.3681639   .3828906    -0.96   0.339    -1.128509     .392181
             |
      res_y4 |
         L1. |  -.3023937   .0756719    -4.00   0.000     -.452663   -.1521243
             |
       _cons |  -.5157553    3.38281    -0.15   0.879    -7.233346    6.201835
------------------------------------------------------------------------------
```

The model might be over-parameterised as the coefficients of the lag 2 variables are insignificantly different from zero. However, the term on the lagged residuals (res_y4) is -0.31 suggesting that around one-third of the disequilibrium is corrected for within one quarter. Diagnostic testing of the residuals gives:

```
. estat hettest

Breusch-Pagan / Cook-Weisberg test for heteroskedasticity
         Ho: Constant variance
         Variables: fitted values of D.y4

         chi2(1)      =     1.93
         Prob > chi2  =   0.1644

. estat bgodfrey, lags(4)

Breusch-Godfrey LM test for autocorrelation
---------------------------------------------------------------------------
    lags(p)  |          chi2               df                 Prob > chi2
-------------+-------------------------------------------------------------
       4     |          3.466               4                   0.4830
---------------------------------------------------------------------------
                        H0: no serial correlation
```

B Critical values for the ADF test

Extract of MacKinnon's Critical Values

n	Model	Point(%)	ϕ_∞	SE	ϕ_1	ϕ_2
1	No Constant	1	-2.5658	(0.0023)	-1.960	-10.04
	no Trend	5	-1.9393	(0.0008)	-0.398	-0.00
		10	-1.6156	(0.0007)	-0.181	-0.00
1	Constant,	1	-3.4336	(0.0024)	-5.999	-29.25
	no trend	5	-2.8621	(0.0011)	-2.738	-8.36
		10	-2.5671	(0.0009)	-1.438	-4.48
1	Constant	1	-3.9638	(0.0019)	-8.353	-47.44
	+ trend	5	-3.4126	(0.0012)	-4.039	-17.83
		10	-3.1279	(0.0009)	-2.418	-7.58
2	Constant,	1	-3.9001	(0.0022)	-10.534	-30.03
	no trend	5	-3.3377	(0.0012)	-5.967	-8.98
		10	-3.0462	(0.0009)	-4.069	-5.73
2	Constant	1	-4.3266	(0.0022)	-15.531	-34.03
	+ trend	5	-3.7809	(0.0013)	-9.421	-15.06
		10	-3.4959	(0.0009)	-7.203	-4.01
3	Constant,	1	-4.2981	(0.0023)	-13.790	-46.37
	no trend	5	-3.7429	(0.0012)	-8.352	-13.41
		10	-3.4518	(0.0010)	-6.241	-2.79
3	Constant	1	-4.6676	(0.0022)	-18.492	-49.35
	+ trend	5	-4.1193	(0.0011)	-12.024	-13.13
		10	-3.8344	(0.0009)	-9.188	-4.85
4	Constant,	1	-4.6493	(0.0023)	-17.188	-59.20
	no trend	5	-4.1000	(0.0012)	-10.745	-21.57
		10	-3.8110	(0.0009)	-8.317	-5.19
4	Constant	1	-4.9695	(0.0021)	-22.504	-50.22
	+ trend	5	-4.4294	(0.0012)	-14.501	-19.54
		10	-4.1474	(0.0010)	-11.165	-9.88

$$C(p) = \phi_\infty + \phi_1 T^{-1} + \phi_2 T^{-2}$$

Source: MacKinnon (1991). Reproduced by permission of Oxford University Press.

C Stata Commands for cointegration

```
/* Importing an excel dataset */
import excel "Z:\Downloads\danish_money_demand.xlsx", */
*/ sheet("Sheet1") firstrow
/* var1 has entry like var1=73:1 so split this as 73 and 1 and then construct a
date series */
gen yr1=substr(var1,1,2)
gen year=yr1+1900
gen q=substr(var1,4,1)
encode q, g(quarter)
gen time=yq(year, quarter)
format time %tq
drop var1 yr yr1 year q quarter
tsset time
/* Plot log of real money (lrm), as level and 1st difference  also ACF and PACF of level
and 1st difference */
twoway (tsline lrm) (tsline d.lrm, yaxis(2))
ac lrm
pac lrm
ac d.lrm
pac d.lrm
/* Plot log of real real output (lry), as level and 1st difference  also ACF and PACF of
level and 1st difference */
twoway (tsline lry) (tsline d.lry, yaxis(2))
ac lry
pac lry
ac d.lry
pac d.lry
/* Plot bond rate (ib), as level and 1st difference  also ACF and PACF of level and
1st difference */
twoway (tsline ib) (tsline d.ib, yaxis(2))
ac ib
pac ib
ac d.ib
pac d.ib
/* Plot deposit rate (id), as level and 1st difference  also ACF and PACF of level and
1st difference */
twoway (tsline id) (tsline d.id, yaxis(2))
ac id
```

```
pac id
ac d.id
pac d.id
/* Plot lrm vs lry ib and id */
twoway (tsline lrm) (tsline lry, yaxis(2))
twoway (tsline lrm) (tsline ib, yaxis(2))
twoway (tsline lrm) (tsline id, yaxis(2))
/* ADF test for the series using Model C for lrm and lry and model B and C for ib and id */
dfuller lrm, trend lags(2) regress
dfuller lry, trend lags(1) regress
dfuller ib, lags(1) regress
dfuller ib, trend lags(1) regress
dfuller id, lags(1) regress
dfuller id, trend lags(1) regress
/* Long-run equation for lry and saving residuals */
reg lrm lry ib id
predict res, resid
/* plotting ACF and PACF of residuals */
ac d.res
pac d.res
/* Unit root test for residuals */
dfuller res, lags(1)
/* Dropping id from equation */
reg lrm lry ib
predict res1, resid
ac d.res1
pac d.res1
dfuller res1, lags(1)
/* Dynamic model for and solving for long-run residuals */
reg lrm lry ib id l.lrm l.lry
gen lrres=lrm-(_b[_cons]+(_b[lry]+_b[l.lry])*lry+_b[ib]*ib+_b[id]*id)/(1-_b[l.lrm])
ac d.lrres
pac d.lrres
dfuller lrres
/* ECM using long-run residuals */
reg d.lrm d.l.lrm d.lry d.l.lry d.ib d.l.ib d.id d.l.id l.lrres
/* Testing zero coefficients on a series of variables */
test d.l.lry d.l.ib d.id d.l.id
/* A restricted ECM model */
```

```
reg d.lrm d.l.lrm d.lry d.ib l.lrres
/* diagnostic tests for ECM */
estat bgodfrey, lag(1)
estat hettest
```

D R Commands for cointegration

```
# Importing an excel dataset
danish <- read_excel("C:/Downloads/danish_money_demand.xlsx")
# Dropping the date variable from the dataset
danish <- danish %>%
  select(-c(date))
# Create a new date variable
date3 <- seq(as.Date("1974-1-1"), as.Date("1987-7-1"), by = "quarters")
danish_tsq <- xts(danish, order.by=date3)
# Plot log of real money (lrm), as level and 1st difference  also ACF and PACF of level
and 1st difference
# Create the differenced series for each of our series
danish_tsq$dlrm <- 100*diff(danish_tsq$lrm)
danish_tsq$dlry <- 100*diff(danish_tsq$lry)
danish_tsq$dib <- diff(danish_tsq$ib)
danish_tsq$did <- diff(danish_tsq$id)
# As the plot has to be done with ggplot as other options  do not permit two graphs
# with different scales. You need to convert the xts dataset back to a data frame
danish1 <- fortify(danish_tsq)
# Plot both lrm and dlrm - you must scale dlrm to have similar units to lrm and
# then invert the formula for the 2nd axis (need to divide by 20 and add 11.6)
ggplot(danish1, aes(x = Index, y = lrm)) +
  geom_line(aes(color = "lrm")) +
  geom_line(aes(y = dlry/20+11.6, color = "d.lrm")) +
  scale_y_continuous(sec.axis = sec_axis(~.*20-11.6*20, name="d.lrm")) +
  labs(x = "Date", y = "lrm", color = "") +
  scale_color_manual(values = c("red", "blue")) +
  theme(legend.position="bottom") +
  ggtitle("Plot of lrm and d.lrm")
# Plot the ACF and PACF of lrm
acf_lrm <- acf(danish_tsq$lrm)
pacf_lrm <- pacf(danish_tsq$lrm)
# Plot the ACF and PACF of d.lrm, adjust the sample to remove missing observation
acf_dlrm <- acf(subset(diff(danish_tsq$lrm), date3>=as.Date('1974-04-01')))
pacf_dlrm <- pacf(subset(diff(danish_tsq$lrm), date3>=as.Date('1974-04-01')))
# Plot log of real real output (lry), as level and 1st difference
ggplot(danish1, aes(x = Index, y = lry)) +
  geom_line(aes(color = "lry")) +
  geom_line(aes(y = dlry/50+6, color = "d.lry")) +
```

```
  scale_y_continuous(sec.axis = sec_axis(~.*50-6*50, name="d.lry")) +
  labs(x = "Date", y = "lry", color = "") +
  scale_color_manual(values = c("red", "blue")) +
  theme(legend.position="bottom") +
  ggtitle("Plot of lry and d.lry")
# Plot the ACF and PACF of lry
acf_lry <- acf(danish_tsq$lry)
pacf_lry <- pacf(danish_tsq$lry)
# Plot the ACF and PACF of d.lry, adjust the sample to remove missing observation
acf_dlry <- acf(subset(diff(danish_tsq$lry), date3>=as.Date('1974-04-01')))
pacf_dlry <- pacf(subset(diff(danish_tsq$lry), date3>=as.Date('1974-04-01')))
# Plot bond rate (ib), as level and 1st difference
ggplot(danish1, aes(x = Index, y = ib)) +
  geom_line(aes(color = "ib")) +
  geom_line(aes(y = dib+0.15, color = "d.ib")) +
  scale_y_continuous(sec.axis = sec_axis(~.-0.15, name="d.ib")) +
  labs(x = "Date", y = "ib", color = "") +
  scale_color_manual(values = c("red", "blue")) +
  theme(legend.position="bottom") +
  ggtitle("Plot of ib and d.ib")
# Plot the ACF and PACF of ib
acf_ib <- acf(danish_tsq$ib)
pacf_ib <- pacf(danish_tsq$ib)
# Plot the ACF and PACF of d.ib, adjust the sample to remove missing observation
acf_dib <- acf(subset(diff(danish_tsq$ib), date3>=as.Date('1974-04-01')))
pacf_dib <- pacf(subset(diff(danish_tsq$ib), date3>=as.Date('1974-04-01')))
# Plot deposit rate (id), as level and 1st difference
ggplot(danish1, aes(x = Index, y = id)) +
  geom_line(aes(color = "id")) +
  geom_line(aes(y = did+0.1, color = "d.id")) +
  scale_y_continuous(sec.axis = sec_axis(~.-0.1, name="d.id")) +
  labs(x = "Date", y = "id", color = "") +
  scale_color_manual(values = c("red", "blue")) +
  theme(legend.position="bottom")
# Plot the ACF and PACF of id
acf_id <- acf(danish_tsq$id)
pacf_id <- pacf(danish_tsq$id)
# Plot the ACF and PACF of d.id, adjust the sample to remove missing observation
acf_did <- acf(subset(diff(danish_tsq$id), date3>=as.Date('1974-04-01')))
```

```
pacf_did <- pacf(subset(diff(danish_tsq$id), date3>=as.Date('1974-04-01')))
# Plot lrm vs lry
ggplot(danish1, aes(x = Index, y = lrm)) +
  geom_line(aes(color = "lrm")) +
  geom_line(aes(y = lry*2, color = "lry")) +
  scale_y_continuous(sec.axis = sec_axis(~./2, name="lry")) +
  labs(x = "Date", y = "lrm", color = "") +
  scale_color_manual(values = c("red", "blue")) +
  theme(legend.position="bottom") +
  ggtitle("Plot of lrm and lry")
# Plot lrm vs ib
ggplot(danish1, aes(x = Index, y = lrm)) +
  geom_line(aes(color = "lrm")) +
  geom_line(aes(y = ib*5+11, color = "ib")) +
  scale_y_continuous(sec.axis = sec_axis(~./5-11/5, name="ib")) +
  labs(x = "Date", y = "lrm", color = "") +
  scale_color_manual(values = c("red", "blue")) +
  theme(legend.position="bottom") +
  ggtitle("Plot of lrm and ib")
# Plot lrm vs id
ggplot(danish1, aes(x = Index, y = lrm)) +
  geom_line(aes(color = "lrm")) +
  geom_line(aes(y = id*10+11, color = "id")) +
  scale_y_continuous(sec.axis = sec_axis(~./10-11/10, name="id")) +
  labs(x = "Date", y = "lrm", color = "") +
  scale_color_manual(values = c("red", "blue")) +
  theme(legend.position="bottom") +
  ggtitle("Plot of lrm and id")
# ADF test for the series using Model C for lrm and lry and model B and C for ib and id
# In all cases using BIC to set lag length
summary(ur.df(danish_tsq$lrm, type="trend", selectlags = "BIC"))
summary(ur.df(danish_tsq$lry, type="trend", selectlags = "BIC"))
summary(ur.df(danish_tsq$ib, type="trend", selectlags = "BIC"))
summary(ur.df(danish_tsq$ib, type="drift", selectlags = "BIC"))
summary(ur.df(danish_tsq$id, type="trend", selectlags = "BIC"))
summary(ur.df(danish_tsq$id, type="drift", selectlags = "BIC"))
/* Long-run equation for lry and saving residuals */
eg_f1 <- dyn$lm(lrm ~ lry + ib + id, data=danish_tsq)
danish_tsq$eg_res1<-resid(eg_f1)
```

```
# plotting ACF and PACF of residuals
acf(danish_tsq$eg_res1)
pacf(danish_tsq$eg_res1)
# Unit root test for residuals
ur.df(danish_tsq$eg_res1, type="drift", selectlags = "BIC")
# Dropping id from equation
eg_f2 <- dyn$lm(lrm ~ lry + ib, data=danish_tsq)
danish_tsq$eg_res2<-resid(eg_f2)
# plotting ACF and PACF of residuals
acf(danish_tsq$eg_res2)
pacf(danish_tsq$eg_res2)
# Unit root test for residuals
ur.df(danish_tsq$eg_res2, type="drift", selectlags = "BIC")
# Dynamic model for and solving for long-run residuals
eg_f3 <- dyn$lm(lrm ~ lry + ib + id + lag(lry, 1) + lag(lrm, 1), data=danish_tsq)
# Saving th vector of coefficients
b_eg_f3 <- coef(eg_f3)
# Genrating the long-run residuals
danish_tsq$eg_res3 <- danish_tsq$lrm-(b_eg_f3[1]+(b_eg_f3[2]+b_eg_f3[5])*danish_tsq$lry +
b_eg_f3[3]*danish_tsq$ib + b_eg_f3[4]*danish_tsq$id)/(1-b_eg_f3[6])
# Plotting the residuals and the ACF and the PACF
autoplot(danish_tsq$eg_res3)
acf(danish_tsq$eg_res3)
pacf(danish_tsq$eg_res3)
# Unit root test on the residuals
ur.df(danish_tsq$eg_res3, type="drift", selectlags = "BIC")
# Estimating an ECM using eg_res3
eg_f4 <- dyn$lm(dlrm ~ dlry + lag(dlry, 1) + dib + lag(did, 1) + did +
lag(did, 1) + lag(dlrm, 1) + lag(eg_res3, 1),
data = danish_tsq)
# As the model looks overparameteriserd we remove some variables
eg_f5 <- dyn$lm(dlrm ~ dlry + dib + lag(dlrm, 1) + lag(eg_res3, 1),
data = danish_tsq)
# Breush-Godfrey test
bgtest(eg_f5, order=1, data=danish_tsq)
# Breush-Pagan test
bptest(eg_f5)
```

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